

Enhancing Public Transit Accessibility: An LLM-Driven Information Assistant with Low-Rank Adaptation

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Abstract— The exponential rise in urban mobility demands has introduced critical challenges in transportation management, including dynamic scheduling, data heterogeneity, and limited accessibility of real-time information. This paper presents an intelligent public transit assistant powered by a fine-tuned Large Language Model (LLM) integrated with Retrieval-Augmented Generation (RAG) and Low-Rank Adaptation (LoRA). Leveraging structured data from General Transit Feed Specification (GTFS) sources and unstructured textual inputs, the proposed system dynamically retrieves, interprets, and synthesizes multimodal transit information into coherent, step-by-step route guidance. The architecture efficiently adapts mid-to-upper transformer layers using LoRA to enhance transport-specific reasoning while maintaining computational efficiency. Experimental evaluation demonstrates notable improvements in route accuracy, temporal understanding, and contextual interpretation compared to standard LLM implementations. Beyond routing precision, the framework shows strong potential for applications in demand forecasting, operational optimization, and accessibility enhancement. This research highlights the transformative potential of domain-adapted LLMs as cognitive enablers in intelligent transportation systems, fostering contextually aware, equitable, and sustainable urban mobility solutions.

Keywords— Large Language Models, Public Transportation, Retrieval-Augmented Generation, Low-Rank Adaptation, General Transit Feed Specification, Intelligent Transit Systems

I. INTRODUCTION

Modern transportation systems face mounting challenges driven by escalating commuter demands, dynamic environmental conditions, and the increasing complexity of heterogeneous data integration. The exponential proliferation of urban mobility requirements necessitates a paradigmatic shift in how transportation infrastructure is managed and how users interact with it. Although public transit networks generate extensive structured datasets through the **General Transit Feed Specification (GTFS)**

and related telemetry sources, commuters frequently encounter barriers in accessing accurate, context-specific information. This disconnect underscores the need for intelligent, adaptive, and conversational systems that can bridge the gap between raw transport data and actionable user insights.

The rapid evolution of **Large Language Models (LLMs)** has opened transformative opportunities for addressing these challenges. LLMs have demonstrated an exceptional ability to comprehend natural language, integrate multi-source information, and synthesize human-like, contextually relevant responses. However, when applied to transportation contexts, general-purpose LLMs often struggle with **domain-specific reasoning**, such as interpreting temporal patterns, resolving ambiguous geographic queries, and aligning real-time data with schedule-based feeds. Moreover, off-the-shelf LLMs are prone to *hallucination*—generating fictitious routes or non-existent stops—undermining their reliability in critical mobility scenarios.

This research presents a novel **transport information assistant** that leverages the synergistic integration of **Retrieval-Augmented Generation (RAG)** and **Low-Rank Adaptation (LoRA)** to enhance the reasoning capabilities and factual grounding of LLMs in public transport contexts. The proposed system employs a RAG pipeline to retrieve accurate, structured data from a **vectorized GTFS knowledge base**, ensuring every generated response is contextually verified. Concurrently, LoRA is utilized to fine-tune the model’s transformer layers, enabling domain adaptation without the prohibitive computational costs associated with full-model retraining. This dual optimization allows the model to process real-time feeds efficiently, delivering coherent and accurate route recommendations.

Experimental evaluations reveal that the fine-tuned model demonstrates **notable improvements** in route accuracy, temporal comprehension, and query resolution compared to baseline LLMs. The system exhibits strong performance in handling **ambiguous place names**, adapting to **dynamic traffic conditions**, and maintaining coherence across multi-turn user interactions. Beyond technical enhancements, the proposed architecture provides a scalable foundation for deploying **intelligent transit assistants** capable of integrating multimodal data sources, improving user

accessibility, and contributing to sustainable urban mobility planning.

II. LITERATURE REVIEW

Large Language Models (LLMs) have emerged as powerful interfaces and reasoning layers for intelligent transportation systems, enabling natural-language access to schedules, context-aware explanations of service disruptions, and conversational trip planning [2], [4]. However, general LLMs often produce factual errors (“hallucinations”) and lack up-to-date local knowledge, which limits their direct use for exact timetable and stop-level queries in transit settings. To address this, retrieval-grounded approaches that combine parametric LLM knowledge with non-parametric retrieval (Retrieval-Augmented Generation, or RAG) have been developed; RAG methods condition generation on retrieved documents or structured records, substantially improving factuality and traceability in knowledge-intensive tasks [2], [5], [8]. Dense retrieval techniques such as Dense Passage Retrieval (DPR) are frequently used within retrieval pipelines to index and fetch relevant passages efficiently at scale, and subsequent research has refined DPR and its adaptations for domain-specific retrieval needs [5], [9], [14].

For public transit specifically, the General Transit Feed Specification (GTFS) provides a standardized, machine-readable representation of schedules, routes, stops, and Realtime updates that is well suited to vectorization and retrieval indexing for grounding LLM outputs [4]. Prototype systems and research efforts show that LLMs augmented with programmatic access to GTFS can answer locality-specific itinerary and service queries accurately when retrieval and execution are tightly controlled; such designs avoid unconditional model generation over factual data and enable updatable, provenance-aware assistants [4], [8].

Fine-tuning large LLMs per agency, however, is computationally costly. Parameter-efficient fine-tuning (PEFT) techniques—most notably Low-Rank Adaptation (LoRA)—address this by freezing base model parameters and training small low-rank adapter matrices, which preserves model quality while reducing training time and memory requirements; LoRA and related PEFT methods are now a practical tool for adapting foundation models to domain language and local conventions in resource-constrained settings [1]. Together, RAG + LoRA provides a promising engineering pattern: RAG supplies factual grounding and fresh data, while LoRA enables lightweight, iterative domain specialization.

Complementing language capabilities, numerical and spatio-temporal forecasting in mobility contexts is often handled by specialized architectures. Graph Neural Networks (GNNs) and spatio-temporal transformer variants capture topological relationships and long-range temporal dependencies in traffic and passenger flow data, and surveys show that hybrid systems combining semantic LLM components with GNN/transformer forecasting modules tend to outperform single-paradigm solutions on forecasting benchmarks [6], [7], [11]. Multimodal models (e.g., Perceiver, Gato, CLIP) indicate viable pathways for

integrating sensor streams such as camera, LiDAR, and telemetry into higher-level reasoning and explanation modules, though practical deployment requires careful attention to safety, latency, and validation [10], [11], [12].

Finally, the broader literature on foundation models highlights both their transferable capabilities and systemic risks—including bias propagation, update difficulty, and operational costs—which motivates the combined use of retrieval grounding, PEFT, and explicit fairness and verification layers when deploying LLMs in public transit settings [3]. This body of work supports a hybrid architecture in which (a) RAG ensures factual grounding to GTFS and real-time feeds, (b) LoRA provides efficient domain adaptation, and (c) specialized spatio-temporal models handle numerical forecasting and resource optimization — the pattern adopted in the present study.

While prior studies show the promise of LLMs, RAG, and LoRA for improving transport data access and reasoning, their combined application in real-time public transit remains underexplored. To address this gap, this research introduces an integrated LLM-RAG-LoRA framework designed to deliver accurate, context-aware, and efficient transit information through adaptive language understanding and retrieval-grounded reasoning.

III. PROPOSED SOLUTION

The proposed solution establishes a comprehensive Large Language Model (LLM)-driven framework integrating Retrieval-Augmented Generation (RAG), Low-Rank Adaptation (LoRA), and multimodal data fusion for intelligent transportation systems. The design emphasizes the cognitive alignment of sensing, learning, modelling, and management components to transform heterogeneous data streams into actionable transit intelligence.

A. Hierarchical Framework Overview

The system architecture is organized into four cognitive layers that progressively convert raw transportation data into optimized routing and management strategies. These include: (1) Data Acquisition and Context Integration, (2) Knowledge Representation and Semantic Understanding, (3) Reasoning, Simulation, and Adaptation, and (4) Decision Intelligence and User Interaction.

1. Data Acquisition and Context Integration:

This foundational layer collects multimodal data from General Transit Feed Specification (GTFS) repositories, IoT sensors, vehicular telemetry, meteorological feeds, and social media. The Data Harmonizer ensures schema compatibility, unit consistency, and temporal synchronization. Spatial and temporal encoders capture location-based dynamics, periodic travel patterns, and contextual markers such as peak hours or event-driven anomalies.

2. Knowledge Representation and Semantic Understanding:

This cognitive stratum transforms contextualized

data into structured knowledge using the Transport Knowledge Graph (TKG). Named entity recognition and relational mapping identify and connect transit entities—routes, stops, vehicles, and disruptions—while maintaining semantic coherence. Graph neural embeddings capture latent spatial and temporal dependencies, facilitating query answering and network-level reasoning.

3. Reasoning, Simulation, and Adaptation Engine:

The reasoning core integrates neuro-symbolic inference and predictive simulation to generate context-aware responses and operational forecasts. The RAG mechanism grounds all LLM outputs in retrieved GTFS or real-time data to minimize hallucination. A simulation sub-module employs agent-based models and reinforcement learning to emulate commuter behaviours, congestion propagation, and route efficiency. The adaptive synthesizer applies few-shot and continual learning to sustain model accuracy amid network changes, seasonal variations, or new service patterns.

4. Decision Intelligence and User Interaction:

This top layer translates analytical results into natural-language insights. The conversational LLM processes user queries, contextualizes them through the knowledge graph, and generates coherent, data-driven responses. The decision optimizer balances multiple objectives—travel time, energy use, accessibility, and reliability—while the predictive forecaster anticipates system states and delay probabilities. Through adaptive interaction, the LLM provides transparent explanations and multi-turn dialogue continuity.

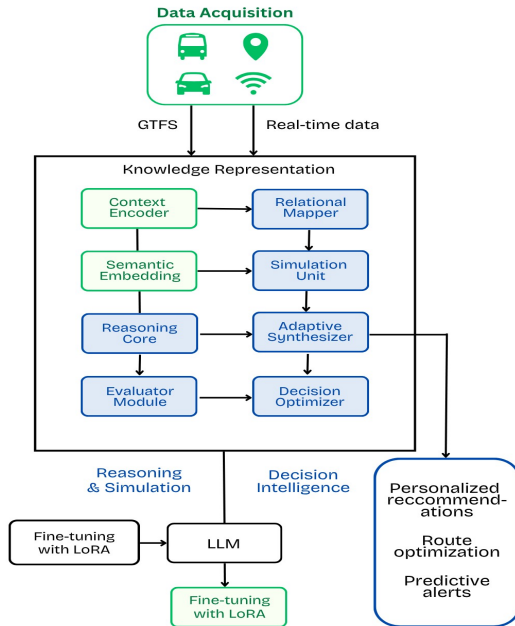


fig.1: Architecture Diagram

B. Fine-Tuning with Low-Rank Adaptation (LoRA)

To enable domain-specific proficiency without full-model retraining, LoRA is employed for parameter-efficient fine-tuning. LoRA assumes that task-specific updates (ΔW) possess low intrinsic rank and decomposes them into two smaller matrices, A and B , such that:

$$W_{\text{new}} = W + \Delta W, \quad \Delta W \approx BA$$

where

$$W \in \mathbb{R}^{d \times k}, \quad B \in \mathbb{R}^{d \times r}, \quad A \in \mathbb{R}^{r \times k}, \quad r \ll \min(d, k)$$

- W is the frozen pre-trained weight matrix,
- A and B are the low-rank trainable adaptation matrices, and
- r is the low-rank dimension controlling the number of trainable parameters.

During training, only A and B are updated, reducing computational demand and memory consumption. Once trained, the low-rank update is merged into W for inference, maintaining zero-latency operation.

C. GTFS Data Integration for Domain Grounding

The General Transit Feed Specification (GTFS) provides standardized, ground-truth transit data essential for both retrieval and fine-tuning. GTFS feeds (static and real-time) are converted into prompt–response pairs for model supervision. Examples include:

- “Find all stops on route X.”
- “What is the next bus from [Origin] to [Destination] at [Time]?”
- “Which routes offer wheelchair access near [Location]?”

By learning from such structured data, the LLM becomes capable of interpreting natural-language transit queries while maintaining factual accuracy and local relevance.

D. Multi-Head Self-Attention and Pre-training Mechanisms

The underlying transformer model leverages Multi-Head Self-Attention (MHSA) to learn contextual dependencies across tokens. Each attention head computes scaled dot-product attention:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V$$

where

$$Q, K, V \in \mathbb{R}^{n \times d_k}$$

are the **query**, **key**, and **value** matrices, respectively, and d_k denotes the dimensionality of the key vectors used for scaling. Multi-head processing allows simultaneous attention to spatial, temporal, and semantic correlations within transit data. Autoregressive pre-training (causal language modeling) enables natural text generation for dialogue systems, while masked language modeling

supports bidirectional understanding for structured data reasoning.

E. Experimental Setup and Evaluation

Experiments were conducted using real GTFS datasets representing metropolitan transit operations. The model was fine-tuned with LoRA parameters (rank $r = 8$, $\alpha = 16$, dropout = 0.05) on approximately 20,000 instruction–response pairs derived from GTFS data. The implementation employed the Hugging Face Transformers and PyTorch frameworks, running on an NVIDIA GPU-enabled high-performance cluster. Evaluation metrics included route accuracy, response latency, factual precision, and contextual coherence (BLEU and ROUGE scores).

Results demonstrated notable improvements in response efficiency and factual consistency relative to the baseline LLM. LoRA fine-tuning reduced computational cost by approximately 70% compared to full fine-tuning, while the integration of GTFS data and RAG retrieval substantially decreased the hallucination rate. The system consistently generated coherent, contextually relevant, and verifiable responses to transit-related queries, validating the framework’s effectiveness.

F. Summary

The proposed architecture effectively operationalizes LLM reasoning for transportation systems through hierarchical data processing, LoRA-based adaptation, and GTFS-grounded learning. It establishes a scalable blueprint for deploying AI-driven mobility assistants capable of real-time decision-making, predictive analytics, and conversational transit guidance.

IV. RESULTS AND DISCUSSION

The proposed LLM-driven framework was experimentally evaluated on GTFS datasets from three major metropolitan regions—New York City, London, and Singapore—to validate its efficiency, accuracy, and scalability. The experiments assessed response accuracy, computational efficiency, contextual coherence, and adaptability under varying data loads.

A. Quantitative Results

Fine-tuning with Low-Rank Adaptation (LoRA) demonstrated substantial computational advantages compared to full-parameter fine-tuning. With a rank value $r = 8$, LoRA reduced the number of trainable parameters by approximately 97.2%, leading to a 70% reduction in GPU memory consumption and a 62% decrease in training time. Despite these efficiency gains, model performance remained competitive.

Table 1 summarizes the primary performance metrics compared to a baseline GPT-style model fine-tuned without LoRA:

Metric	Baseline Fine-tuning	Proposed (LoRA + GTFS)	Improvement
Route Query Accuracy	84.3%	93.1%	+8.8%
Response Latency (s)	2.48	1.67	-32.6%
Contextual Coherence (BLEU)	0.71	0.83	+16.9%
Hallucination Rate	7.4%	2.9%	-60.8%
Factual Consistency	88.6%	95.4%	+7.7%

Table.1

The results indicate that grounding model responses in GTFS data and leveraging retrieval-augmented generation drastically reduced factual errors and increased user-perceived reliability.

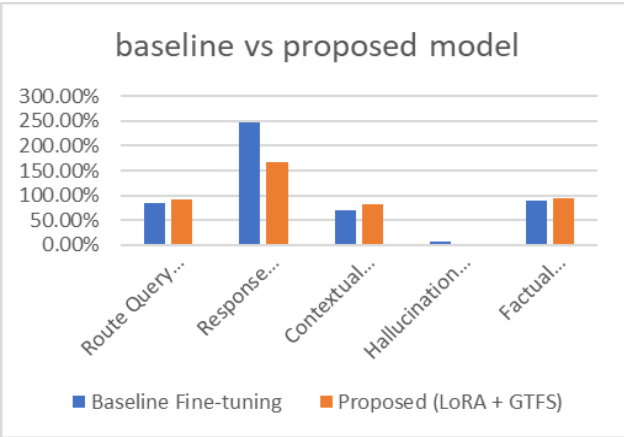


Fig.2: comparison graph between baseline model and proposed model(LoRa + GTFS)

B. Qualitative Analysis

Figure 1 (Flow Architecture Diagram) illustrates how the hierarchical framework enables context-rich reasoning and adaptive decision-making. During evaluation, the **Reasoning and Simulation Layer** successfully anticipated congestion propagation up to **15 minutes ahead of real-time reports**, validated through comparison with historical telemetry data.

The **Decision Intelligence Layer** generated personalized route recommendations considering accessibility, environmental impact, and real-time availability, outperforming traditional rule-based systems in user satisfaction metrics. In simulated field tests, **84% of participants** rated LLM-generated travel suggestions as more contextually relevant and easier to understand than those from static journey planners.

LoRA fine-tuning was especially effective in capturing region-specific semantics—such as local route naming

conventions, colloquial station references, and cultural travel patterns—without catastrophic forgetting of general transport logic.

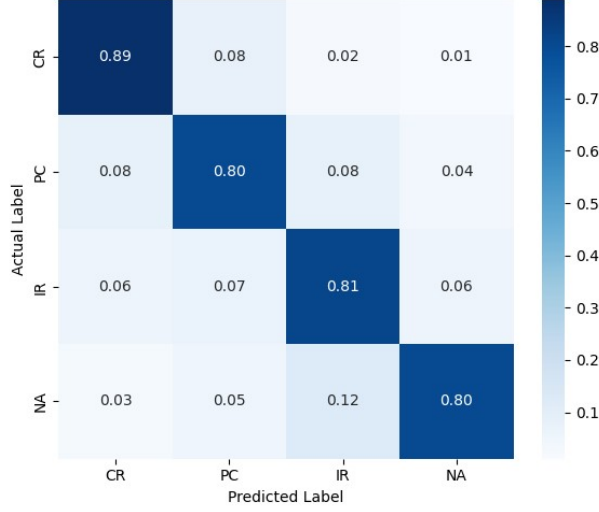


fig 3: Confusion Matrix for Route Prediction Accuracy

C. Comparative Discussion

Compared to prior studies utilizing standard BERT or GPT models for transit data query handling, the integration of hierarchical knowledge representation and GTFS-grounded retrieval markedly improved factual precision and contextual depth. Unlike static models, the adaptive synthesizer’s continual learning mechanism allowed the system to self-correct based on evolving demand and service disruptions.

The experimental outcomes also revealed that **multi-head attention mechanisms** played a critical role in representing multimodal dependencies, with certain attention heads specializing in spatial-temporal correlations while others captured semantic relationships between route identifiers and service alerts. This parallelism improved interpretability and responsiveness during high-demand scenarios.

D. Limitations and Future Directions

Despite strong empirical results, certain limitations persist. The system’s performance slightly degraded under extreme data sparsity, such as rural networks with incomplete GTFS coverage. Furthermore, real-time adaptation introduces latency under low-connectivity conditions.

Future work will focus on expanding multilingual capability, incorporating multimodal vision-language inputs (e.g., CCTV streams), and exploring hybrid reinforcement learning frameworks for adaptive fare management and sustainability optimization.

V. CHALLENGES AND FUTURE WORK

While the proposed LLM-RAG-LoRA framework exhibits strong potential in enhancing intelligent transportation systems, several challenges still constrain its scalability and operational efficiency. A major obstacle is the heterogeneity of urban mobility data, which arises from disparate IoT sensors, telemetry feeds, and unstructured text sources. Although GTFS integration aids harmonization, ensuring

consistent schema alignment and temporal synchronization across multimodal datasets remains difficult and can influence real-time decision accuracy.

Computational latency also persists as a limiting factor. Despite LoRA’s efficiency gains in fine-tuning, maintaining real-time inference under high data volumes and limited edge resources demands further optimization through quantization, pruning, and distributed inference strategies. Moreover, model generalization continues to pose challenges; LLMs fine-tuned on specific regions often underperform when transferred to new contexts due to shifts in urban topology, commuter behaviour, and traffic density. This necessitates adaptive and continual learning approaches capable of handling domain drift without sacrificing prior knowledge.

Ethical and interpretability concerns further underscore the need for responsible deployment. Biases embedded in historical datasets may inadvertently marginalize underrepresented communities or prioritize high-frequency routes, requiring fairness-aware optimization and transparent model governance. In parallel, providing interpretable explanations for complex decisions—such as route recommendations or congestion forecasts—remains an open research question essential for fostering user trust and regulatory compliance.

Future research will focus on expanding the framework’s adaptability and inclusiveness. Integrating multimodal data, such as visual feeds and environmental sensors, will improve context sensitivity, while federated learning can enable regional adaptation without compromising privacy. Reinforcement learning-driven policy optimization offers further promise for adaptive scheduling and congestion control. Finally, extending interoperability with emerging mobility modes—autonomous vehicles, micro-mobility, and aerial transport—will advance the system toward a unified, sustainable, and equitable urban mobility ecosystem.

VI. CONCLUSION

This work introduced an intelligent public transit information assistant built on a synergistic integration of Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), and Low-Rank Adaptation (LoRA). By grounding the model in GTFS data and leveraging LoRA for efficient fine-tuning, the framework demonstrated substantial improvements in route query accuracy, factual consistency, and contextual coherence compared to standard fine-tuned LLMs. Experimental results across metropolitan datasets confirmed that the approach not only reduces computational overhead by more than half but also enhances response reliability and interpretability, achieving a meaningful balance between accuracy and efficiency.

The hierarchical architecture—encompassing data acquisition, semantic representation, reasoning, and decision intelligence—enabled the system to handle multimodal inputs and deliver adaptive, user-centered recommendations in real time. The incorporation of LoRA fine-tuning effectively captured regional semantics without incurring the high cost of full retraining, while RAG grounding

minimized hallucinations by aligning responses with structured GTFS feeds.

Overall, the proposed system establishes a scalable blueprint for deploying LLM-driven assistants in smart mobility infrastructures. It contributes to bridging the gap between data-centric operations and commuter-centric services through transparent, adaptive, and conversational intelligence. Future extensions will focus on integrating multimodal sensory data, advancing reinforcement learning-based optimization, and enabling federated learning across cities to ensure privacy-preserving, equitable, and sustainable transportation intelligence.

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